Process or Materials Review Request Form

Request Title: Nickel (Ni) wet etch off semi-clean silicon (Si) wafer in semi-clean wet bench CMOS metal (wbclean3) using dilute piranha (H2SO4+H2O2+H2O) for NiSi/NiSi2 salicide on Si

Requestor: Jie "Jay" Yao **Date: 25th September 2021**

Badger ID: JieYao@WavefrontVision.com

Company name/PI: Wavefront Vision Inc.

Send completed form to: snf-promcommittee@lists.stanford.edu

Hi, Michelle, Uli and Maurice,

May I make the following PROM request, please?

Nickel (Ni) wet etch off semi-clean silicon (Si) wafer in the semi-clean wet bench CMOS metal (wbclean3) using dilute piranha (H2SO4+H2O2+H2O) for NiSi/NiSi2 salicide on Si according to https://web.stanford.edu/class/ee311/NOTES/Silicides.pdf.

Thank you very much in advance!!

-- Jay

1. **The chemical or material.** Nickel (Ni) wet etch off semi-clean silicon (Si) wafer in the semi-clean wet bench CMOS metal (wbclean3) using dilute piranha (H2SO4+H2O2+H2O) for NiSi/NiSi2 salicide on Si.
2. **Vendor/manufacturer info:** SNF standard chemical vendors.
3. **Reason for request:** According to https://web.stanford.edu/class/ee311/NOTES/Silicides.pdf, NiSi/NiSi2 is a standard silicide for Si CMOS processes. We need to wet etch nickel (Ni) off semi-clean silicon (Si) wafer in the semi-clean wet bench CMOS metal (wbclean3) using dilute piranha (H2SO4+H2O2+H2O) as part of the industrial standard NiSi/NiSi2 salicide process for Si CMOS wafers.
4. **Process Flow:** Wet etch nickel (Ni) on semi-clean silicon (Si) wafer in dilute piranha (namely, sulfuric acid (H2SO4) and hydrogen peroxide (H2O2) and water (H2O)) in the semi-clean wet bench CMOS metal (wbclean3) at room temperature without heating.
5. **Amount and form.** n/a.
6. **Storage:** SNF standard sulfuric acid (H2SO4) and hydrogen peroxide (H2O2).
7. **Disposal**: SNF standard AWN via wbclean3.

nickel density = 9 gram/cm^3; nickel thickness < 600 Å;
nickel area = 1 single-sided 4-inch wafers < 80 cm^2;
nickel volume < 5e-4 cm^3; nickel mass < 5 milli-gram;
acid volume > (4-inch)^3 = 1 liter;
one single wafer of nickel in acid < 5 milli-gram/liter as the absolute maximum limit in acid prior to any dilution in the acid/base neutralization system (AWN)!!
As a realistic scenario, half nickel thickness and half a cassette in one gallon acid tank gives approximately 30nm * [π*(5cm)*2/wafer * 12.5wafers] * 8.9g/cm^3 / 3.79L = 7 mg/L in acid prior to any dilution by the acid/base neutralization system (AWN).

To be completed by the PROM COMMITTEE:

Special handling requirements:

On-going approval or staff interaction requirements:

Approved By/Date:

V1.0